

Classification of tooth preparation (cavity preparation)

1. According to tooth surface involvement:

Simple cavity: cavity involve only one surface of the tooth Ex: occlusal cavity (O) CL I.

Compound cavity: cavity that involved two surfaces Ex: mesio-occlusal (MO) CLII.

Complex cavity: cavity involve three or more surfaces.

2. According to site involved:

Site 1: pits, fissure and enamel defects on occlusal surfaces of posterior or other smooth surfaces.

Site 2: proximal enamel in relation to areas in contact with adjacent teeth.

Site 3: the cervical one third of the crown or following gingival recession and the exposed root surface.

3. ICDAS:

The ICDAS uses a two-stage process to record the status of the caries lesion in occlusal surface.

The first is a code for the severity of the caries lesion and the second is for the restorative status of the tooth which determined visually on a scale of 0 to 6:

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0 = sound tooth structure

1 = First visual change in enamel

2 = Distinct visual change in enamel

3 = Enamel breakdown, no dentin visible

4 = Dentinal shadow (not cavitated into dentin)

5 = Distinct cavity with visible dentin

6 = Extensive distinct cavity with visible dentin.

The second is the severity code paired with a restorative/sealant code 0 to 8:

0 = Not sealed or restored

2 = Sealant, partial

3 = Sealant, full; tooth-colored restoration

4 = Amalgam restoration

5 = Stainless steel restoration

6 = Ceramic, gold, PFM (porcelain-fused-to-metal) crown or veneer

7 = Lost or broken restoration

8 = Temporary restoration

4. According to G.V. Black:

CLI cavity: these are cavities of CL I lesions the following surfaces are involved:

- Occlusal pits and fissures of premolars and molars fig 1A.
- Facial & lingual pits and fissures of mandibular molars fig 1B.
- Palatal pits of maxillary incisors, most frequently in the pit near the cingulum fig 1C.

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A

B

C

Fig 1: CI I

CL II restorations: is the extension of CLI restoration into the proximal surfaces of premolars and molars, the following surfaces are involved:

- Two surface restoration of posterior teeth fig 2A.
- Three surface restoration of posterior teeth fig 2B.
- Four surface (or more) restoration of posterior teeth fig 2C.



A



B



C

Fig 2. CI II

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CLIII: affects the interproximal surface of incisors and canines fig 3.



Fig 3: CL III

CLIV: involves a larger surface area, including the incisal edges and interproximal surface of incisors and canine fig 4.



Fig 4: CL IV

CL V: - gingival third of the facial or lingual surfaces of any tooth.

- -Root of a tooth near the cemento-enamel junction fig 5.

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Fig 5: CL V

CI IV: Incisal edge of anterior teeth or on cusp tip of posterior teeth

Steps of tooth preparation (cavity preparation)

Initial Tooth Preparation

Step 1: Initial depth and outline form

Step 2: Primary resistance form

Step 3: Primary retention form

Step 4: Convenience form

Final Tooth Preparation

Step 5: Removal of defective restorative material and/or soft dentin

Step 6: Pulp protection

Step 7: Secondary resistance and retention forms

Step 8: External wall finishing

Step 9: Final procedures: debridement and inspection

Step 1: Initial depth and outline form

Initial depth is the first step in tooth preparation to be established

Outline form is the shape of the cavity regardless of the type of tooth preparation with the following

1. All faults (such as caries) are included.
2. All preparation margins are usually placed in a position that allows inspection and finishing of the subsequent restoration margins.

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Step 2: Primary resistance form

Is the shape and placement of the preparation walls (cavity walls) that enable the remaining tooth structure, as well as the restoration to withstand masticatory forces primarily oriented parallel to the long axis of the tooth without fracture.

- Primary resistance form is obtained through use of a preparation design that conserves as much healthy tooth structure as possible.
- The internal line angles are slightly rounded so as to limit stress concentration in these areas, which increases tooth resistance to fracture.
- The minimal occlusal thickness for restoration, for appropriate resistance to fracture, for amalgam is 1.5 to 2 mm and composite resins have no minimal thickness.

Step 3: Primary retention form

Is the shape of the cavity that permits the restoration to resist displacement through the tipping or lifting force.

- The design of cavity preparation is directly related to the retention needs of the anticipated restorative material.
- Composite restorations are primarily retained in the tooth by micromechanical and, depending on the adhesive, chemical bonding that is established between the restoration and the tooth structure.

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- Amalgam restorations are primarily retained in the tooth by opposing buccal & lingual walls of the cavity should be parallel to each other or converge occlusally (5°) fig 6. Also pupal floor should be flat to prevent restoration movement.

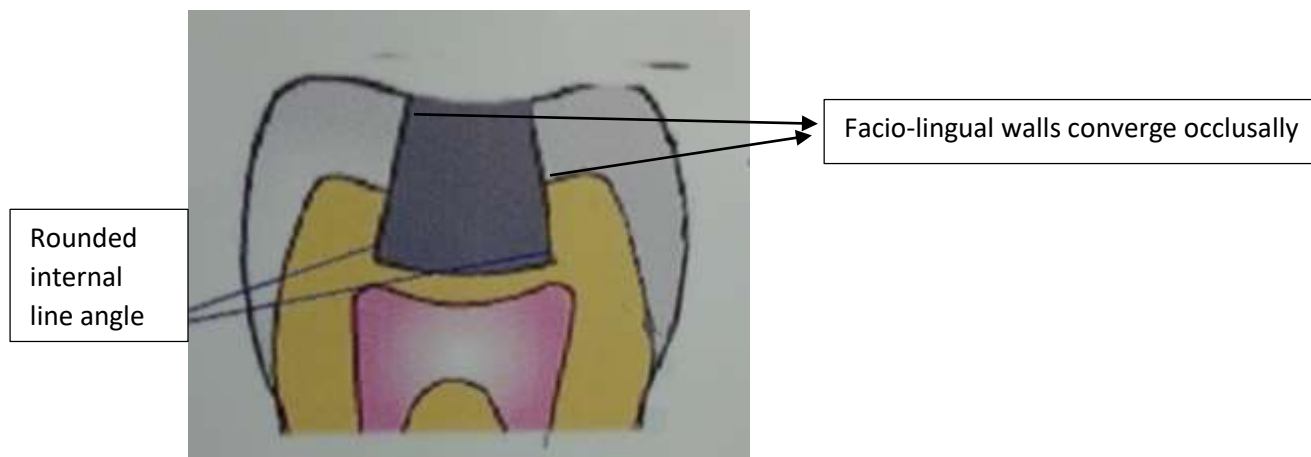


Fig 6: Primary retention forms for amalgam

Step 4. Convenience form:

Convenience form is the shape of the cavity that provides adequate observation, accessibility, and ease in the preparation and restoration of the tooth.

Step 5. Removal of remaining caries

Deep dentinal caries can be removed by using spoon excavator or large round bur with slow speed hand piece.

Step 6: Pulp Protection

Composed of deep dentin + liner/base between the restoration and the pulp.

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- Deep dentin provide protection from

- (1) Heat generated by rotary instruments during subsequent steps,
- (2) Noxious ingredients of various restorative materials,
- (3) Thermal changes conducted through restorative materials,
- (4) Forces transmitted through materials to the dentin,
- (5) Galvanic shock, and
- (6) The ingress of bacteria and/or noxious bacterial toxins through microleakage.

The term *liners* used in a relatively thin film while *base* is used to describe the placement of materials, used in thicker dimensions, beneath permanent restorations to provide for mechanical, chemical, and thermal protection of the pulp.

Generally, it is desirable to have approximately a 2-mm dimension of bulk between the pulp and amalgam restorative material. This bulk may include remaining dentin, liner, or base.

Composite resin materials, which are thermal insulators, do not require the same bulk of material (dentin + liner/base) between the restoration and the pulp.

Step 7: Secondary retention and resistance forms

The secondary retention and resistance forms are of two types:

- (1) Mechanical preparation features (retentive grooves and coves).
- (2) Treatments of the preparation walls with etching, priming, and adhesive materials.

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Step 8: External Wall Finishing

Involve making the wall smooth and removing of unsupported enamel and placement of bevel on the cavosurface margin such that the anticipated restorative material has the greatest likelihood of clinical success.

Step 9: Final Procedures: Debridement and Inspection

Debridement (cleaning) of the tooth preparation involves use of the air/water syringe to remove visible debris with water.

Inspection of the preparation so as to ensure adherence to all principles of preparation design.